Chapter 1

Introduction to Data Visualization

It is difficult for the human mind to look at a list of numbers and identify the patterns in them, so we often use these numbers to make a picture. These pictures are called *graphs*, *charts*, or *plots*. Often, the right picture can make the meaning in the data obvious. **Data visualization** is the process of making pictures from numbers.

1.1 Types of Variables

Depending on the type of data and what you are trying to demonstrate about it, you will use different types of data visualizations. Before we get to the more complex types of data visualizations, let's look at the two types of variables we are usually looking to visualize: *categorical* and *numerical*.

Categorical or qualitative variables are those that can be put into categories. For example, the type of shoes a person wears (sneakers, sandals, etc.) is a categorical variable. The type of shoes sold is a categorical variable. The name of the salesperson (John, Jane, etc.) is also a categorical variable.

Numerical or quantitative variables are those that can be measured with numbers. For example, the number of boxes of cookies sold is a quantitative variable. The average temperature on a given day is also a quantitative variable. Quantitative variables can be further divided into *discrete* and *continuous* variables. Discrete variables are those that can only take on specific values, such as the number of pencils sold (you can't sell 3.5 pencils). Continuous variables can take on any value within a range, such as the average height of students in a class (which can be 5.5 feet, 1.65 m, etc.).

Further, we can also classify data into univariate, bivariate, and multivariate data. Univariate data has only one variable, such as the number of boxes of cookies sold. Bivariate data has two variables, such as the average temperature and total sales for a lemonade stand. Multivariate data has more than two variables, such as the average temperature, total sales, and type of ookie sold. We will only look at univariate and bivariate data!

1.2 How do we describe data?

There are different descriptive methods for representing data, and they can usually be grouped into three categories: *tabular*, *graphical*, and *numerical*.

Tabular methods include tables, which are a way of organizing data into rows and columns. They are useful for displaying raw data and making it easy to look up specific values. However, they can be difficult to interpret when dealing with large datasets or when trying to identify patterns.

Graphical methods include bar charts, line graphs, pie charts, and scatter plots. They are useful for visualizing data and making it easier to identify patterns and trends.

Numerical methods include measures of central tendency (mean, median, mode) and measures of variability (range, variance, standard deviation). They are useful for summarizing data and providing a numerical representation of the data's characteristics.

We will focus on graphical methods in this chapter, but we will also touch on tabular and numerical methods in later chapters.

1.3 Univariate Data

We will start with univariate data, which has only one variable. For example, the number of boxes of cookies sold by each salesperson is univariate data. Bar charts, line graphs, and pie charts are all good for visualizing univariate, *categorical* data. We will explore these types of graphs in the next sections.

1.3.1 Bar Chart

Bar charts are typically used to show the values of a categorical variable. Each bar represents a category, and the height of the bar represents the value of that category. Bar charts can have vertical or horizontal bars to represent this data.

Here is an example of a bar chart.

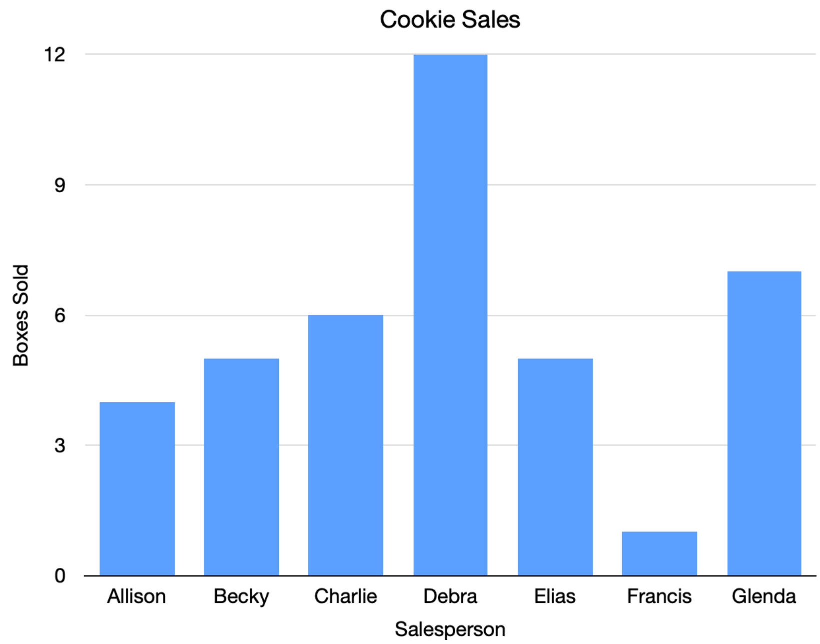

Figure 1.1: A bar chart showing cookie sales per person.

Each bar represents the cookie sales of one person. For example, Charlie has sold 6 boxes of cookies, so the bar goes over Charlie's name and reaches to the number 6.

Looking at this chart, you probably think, "Wow, Debra has sold a lot more cookies than anyone else, and Francis has sold a lot fewer."

The same data could be in a table like this:

Salesperson	Boxes Sold
Allison	4
Becky	5
Charlie	6
Debra	12
Elias	5
Francis	1
Glenda	7

A table (especially a large table) is often just a bunch of numbers. A chart helps our brains understand what the numbers mean.

Bar charts can also go horizontally.

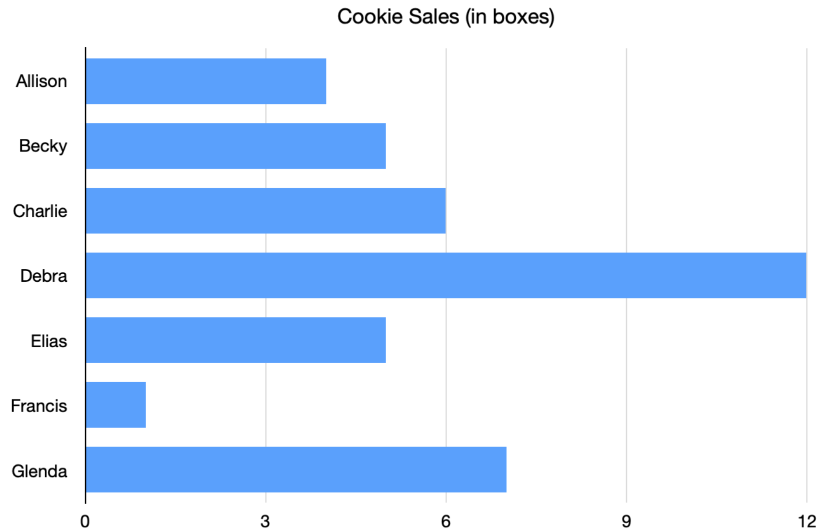

Figure 1.2: A horizontal bar chart showing the same cookie sales data.

Sometimes we use colors to explain what contributed to the number.

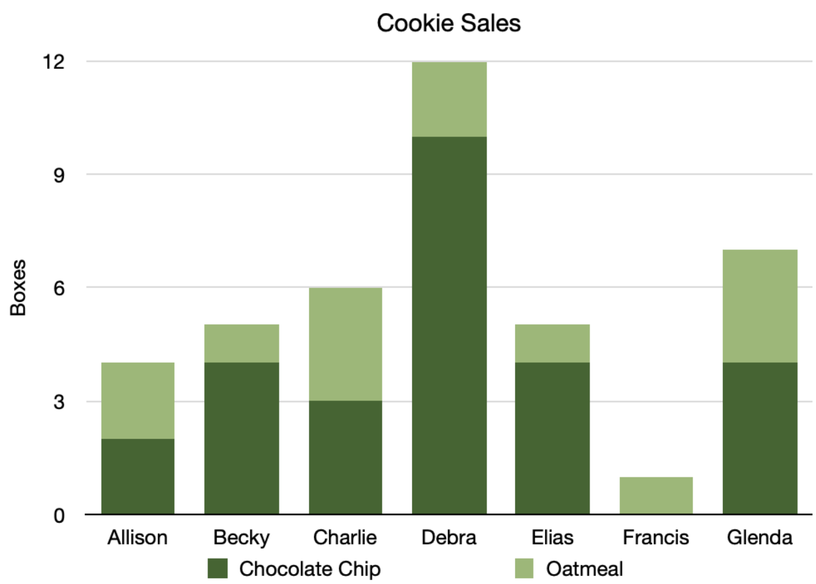

Figure 1.3: A bar chart with different colors showing types of cookies.

This tells us that Becky sold more boxes of chocolate chip cookies than boxes of oatmeal cookies.

Make Bar Graph

Go back to your compound interest spreadsheet and make a bar graph that shows both balances over time:

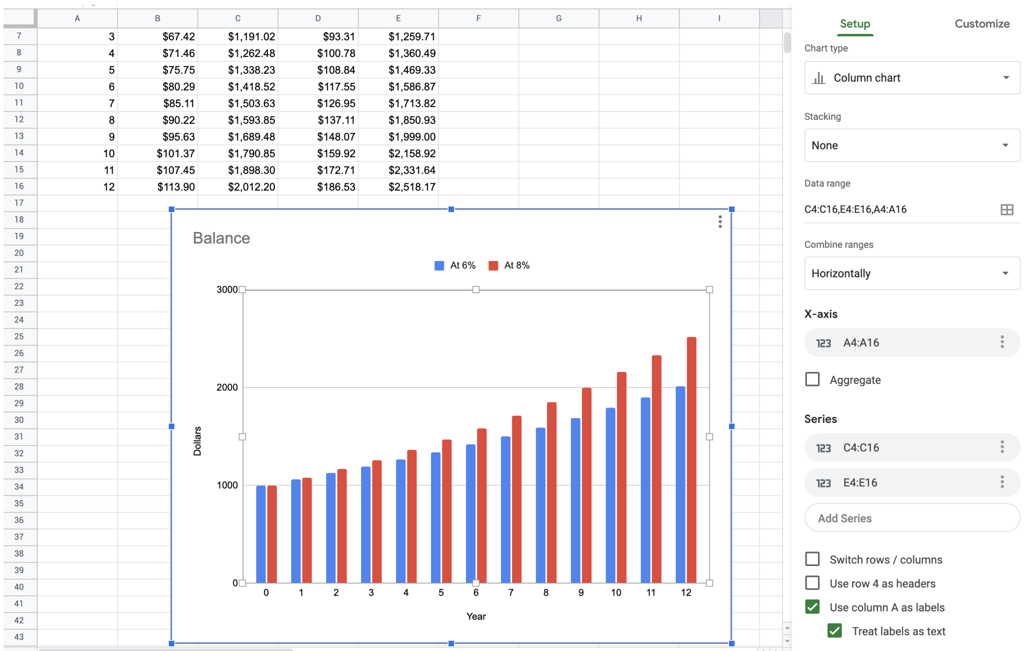

Figure 1.4: A bar graph made in google sheets showing interest.

The year column should be used as the x-axis. There are two series of data that come from C4:C16 and E4:E16. Tidy up the titles and legend as much as you like. Looking at the graph, you can see the balances start the same, but balance of the account with the larger interest rate quickly pulls away from the account with the smaller interest rate.

1.3.2 Line Graph

Here is a line graph:

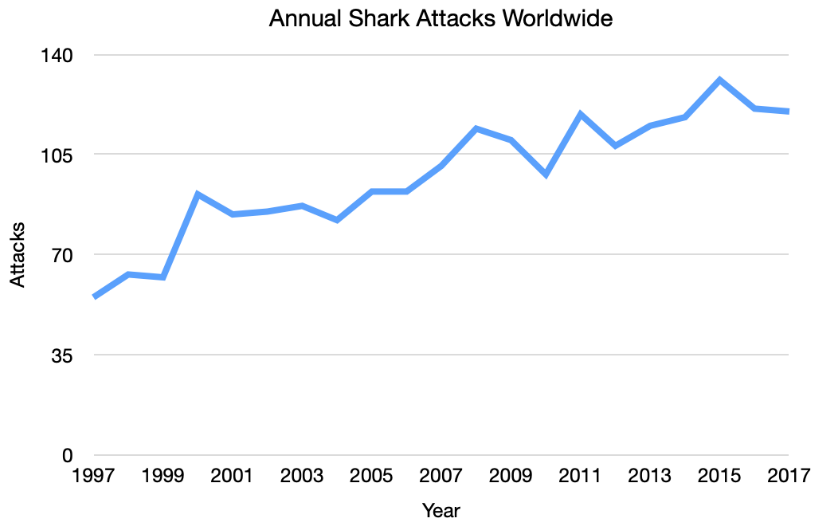

Figure 1.5: A line graph showing shark attacks per year over two decades.

These are often used to show trends over time. Here, for example, you can see that the number of shark attacks has been increasing over time.

You can have more than one line on a graph.

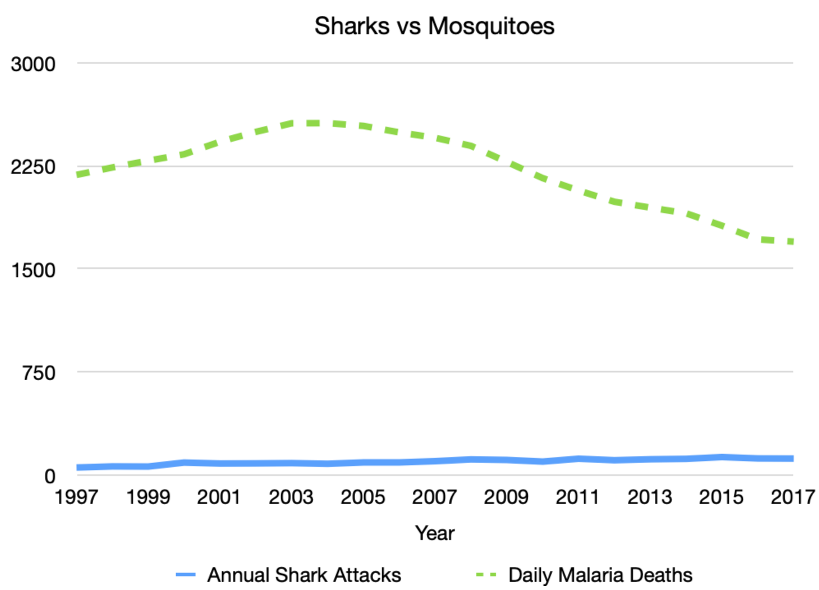

Figure 1.6: A line graph showing shark deaths versus mosquito deaths.

1.3.3 Pie Chart

You use a pie chart when you are looking at the comparative size of numbers. This is best for comparing percentages of a whole that sum to 100%. Here we can see that Nitrogen makes up 78% of the gases in the air.

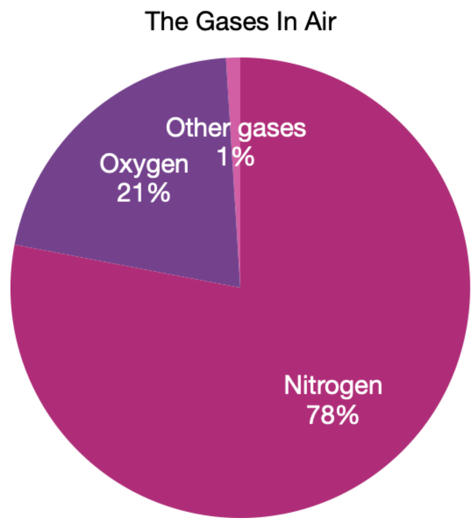

Figure 1.7: A pie chart of the various gases in the air.

When to use a Pie Chart Make a Pie Chart in Python

For *quantitative*, univariate data, we often use dotplots and stemplots to visualize smaller sets of data; and histograms, cumulative frequency plots, and boxplots to visualize larger sets of data. We will explore these types of graphs below.

1.3.4 Dotplot

A *dotplot* is one of the simplest ways to display single, quantitaive variables. Each observation is represented by a dot placed above its value on a number line. If a value appears more than once, the dots stack vertically.

Dotplots are most effective for small to moderate data sets. If the data set is too large, a dotplot can become cluttered and hard to read. In that case, a histogram or a boxplot may be a better choice.

When to Use a Dotplot

- You have 1 variable.

- The variable is quantitative.
- You want to see the distribution and the individual data values.
- The data set is not too large.

How to Make a Dotplot

1. Draw a horizontal number line (x-axis) that covers the data range.
2. Choose a scale that fits the entire range of values.
3. Place one dot above the location of each observation.
4. If multiple observations have the same value, stack the dots vertically.

Here is an example of a dotplot.

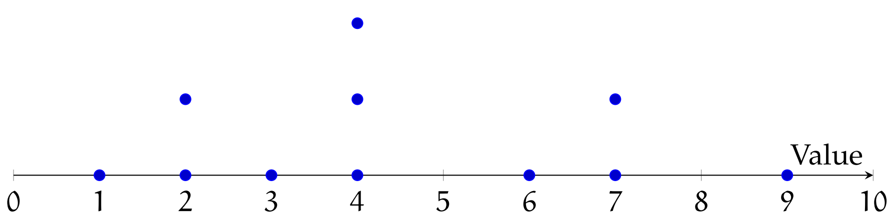

Figure 1.8: A dotplot showing a small sample distribution.

How to Read a Dotplot

Each dot shows the location of one data value. To find the value of a dot, look straight down to the number line.

A dotplot helps you describe the distribution by answering questions like:

- Spread: How spread out are the values (range and clustering)?
- Shape: Do the dots form a roughly symmetric shape, or is the distribution skewed?
- Center: Where is the distribution "balanced" (a typical value)?
- Outliers: Are there values far away from the rest?

Sometimes a dotplot shows a distribution with a long tail to one side. For example, salary data often has many values in the lower-to-middle range and a few extremely high values. This creates a right-skewed distribution.

Outliers can strongly affect numerical summaries, especially the mean.

Connecting Outliers and the Mean (Seesaw Idea) A useful way to think about the mean is as a balancing point. Imagine the dotplot as a seesaw: points far from the main cluster have more "pull." So a single extreme value can shift the mean noticeably, even if most data values stay the same.

Comparative Dotplots

A *comparative dotplot* uses the same numerical scale to compare two (or more) groups. This is useful when you want to compare distributions side-by-side.

When comparing groups, focus on differences in:

- center (typical value),
- spread,
- shape (skew/clusters),
- and outliers.

Make a Dotplot in a Spreadsheet

You can create a dotplot in Excel or Google Sheets using a scatter plot and a helper column.

1. Put your data values in column A (starting in A2). Sort them (recommended).
2. In cell B2, enter:
   `=COUNTIF($A$2:A2, A2)`
   Fill down. This creates stacking heights for repeated x-values.
3. Insert a Scatter chart using column A as x-values and column B as y-values.
4. Format the chart:
   - markers only (no lines),
   - hide the y-axis labels (y is only for stacking),
   - add titles and adjust the x-axis scale.

1.3.5 Stemplot

A *stemplot* (also called a *stem-and-leaf plot*) is an effective way to summarize **one quantitative variable** when the data set is not too large. Like a dotplot, it helps you see the overall distribution, but a stemplot has an extra advantage: it still shows the exact data values.

To make a stemplot, each observation is split into two parts:

- the stem, which is the left-most part of the number (leading digit(s))
- the leaf, which is the remaining part, usually the final digit

For example, 23 can be written as 2 | 3.

How to Make a Stemplot

1. Choose stems and leaves. Decide where to split each number into a stem and a leaf. There is no single rule for this. Your goal is to choose a reasonable number of stems.
2. Draw a vertical divider. Stems go on the left; leaves go on the right.
3. List all stems in order. Make sure the stems cover the full range of your data.
4. Write leaves for each observation. Place each leaf next to its correct stem.
5. Sort leaves. Put the leaves for each stem in increasing order.
6. Include a key. The key shows how to interpret the stems and leaves.

Choosing the Number of Stems

If you use *too few stems*, you can hide patterns because the data get lumped together. If you use *too many stems*, you can also hide patterns because the plot becomes too spread out.

A common strategy is to start with stems that represent groups of 10 (tens digits as stems), and then adjust if needed.

Sometimes a single stem represents too wide of an interval. In that case, you can split stems by repeating each stem twice:

- the first row holds leaves 0–4
- the second row holds leaves 5–9

For example, if your data run from 10 to 49, you can use stems

$$1, 1, 2, 2, 3, 3, 4, 4$$

where the first 1 represents 10–14 and the second 1 represents 15–19, and so on.

Example

Suppose these are housing costs (in dollars per month):

$$255, 259, 304, 306, 309, 317, 323, 327, 332, 334, 342, 349, 350, 351, 362, 367, 376, 384, 401, 403$$

If we use the hundreds digit as the stem and the last two digits as the leaf, we can split each stem into two rows: (top row for leaves 00–49, bottom row for leaves 50–99)

Stem	Leaves
2	55 59
2	
3	04 06 09 17 23 27 32 34 42 49
3	50 51 62 67 76 84
4	01 03
4	

Key: 3 | 04 = 304 dollars per month

How to Read a Stemplot

A stemplot gives you several kinds of information at once:

- Exact values: Each leaf represents a single observation.
- Frequencies: The number of leaves in a row tells how many observations fall in that interval. For example, if a stem row has 9 leaves, that means 9 observations are in that range.
- Gaps: If a stem has no leaves, that means there are no observations in that interval.
- Shape and spread: You can see whether the values cluster in certain ranges and whether the distribution is skewed.
- Center: You can estimate a typical value by locating where the leaves cluster most.
- Outliers: Unusually high or low values may stand out because they appear far from the rest.

Just like with dotplots, you can imagine the distribution as something that could "balance" on a finger: the place it would balance is a rough sense of the center.

Make a Stem Plot in a Spreadsheet Make a Stem Plot in Python

1.3.6 Histogram

A *histogram* is one of the most popular ways to display single quantitaive variables. It is especially useful for showing patterns in larger data sets, where dotplots or stemplots can become overwhelming. A histogram is like a stemplot turned on its side: instead of listing individual values, we group values into intervals called classes or bins, and draw a bar for each bin.

Histograms are best when you care about the overall shape of a distribution (clusters, skew, gaps, outliers), not the exact value of every data point.

A histogram can be drawn using frequency (counts), relative frequency (proportions), and percent frequency (percentages). These three versions have the same shape if they use the same bins. The difference is only the y-axis scale.

How to Make a Histogram

1. Create groups from continuous data. This process is called binning.
2. Draw the x-axis and y-axis, and scale them to fit the class intervals and the frequencies (or relative frequencies or percentages).
3. Draw one bar for each class. The height of each bar equals the frequency (or relative frequency or percent) for that class.
4. Draw the bars touching with no gaps (because the data values are continuous and the bins connect).

How to Read a Histogram

- Each bar represents a single class (bin). There is one bar per class.
- The bins are placed on the x-axis in increasing order, like a number line.
- The height of a bar tells how many observations fall in that class (for a frequency histogram).
- A class with no bar means no observations fall in that interval (a gap).

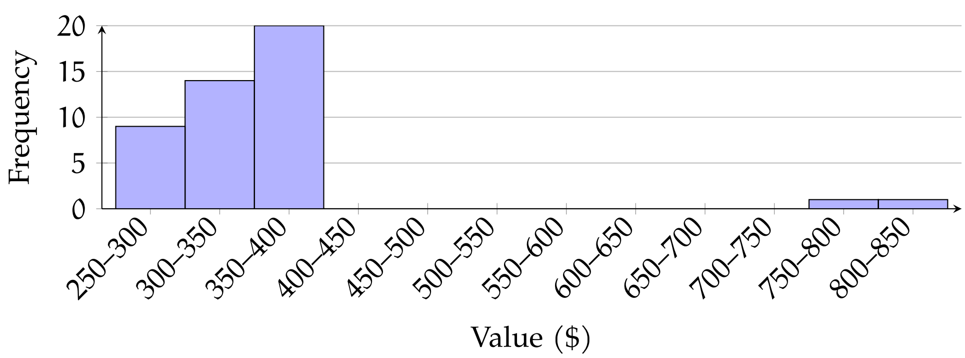

Figure 1.9: A histogram using 50-dollar bins illustrating the example counts (9, 14, 20, ... and 0,1,1 beyond $700).

Example of Interpreting Bars

Suppose a histogram uses the bins \$250–\$300, \$300–\$350, \$350–\$400, and so on.

- If the first bar (250–300) has height 9, then 9 observations are between \$250 and \$300.
- If the second and third bars (300–350 and 350–400) have heights 14 and 20, then $14 + 20 = 34$ observations are between \$300 and \$400.
- If there is no bar for 700–750, then there are 0 observations between \$700 and \$750.
- If the bars beyond 700 have heights 0, 1, and 1, then $0 + 1 + 1 = 2$ observations are at least \$700.

A histogram can help you to describe the distribution by focusing on shape, center, spread, gaps, and outliers.

Making a Histogram in a Spreadsheet

You can make a histogram in Excel or Google Sheets.

1. Put the data in one column.
2. Choose bin widths (for example, bins of width 50: 250–300, 300–350, 350–400, etc.).
3. Insert a histogram chart (or use a column chart after building a frequency table).
4. Check the bin width and the endpoints. Changing the bins can change the appearance of the distribution.

Making a Histogram in Python

Python can generate a histogram quickly and lets you change the bin width easily.

```
import matplotlib.pyplot as plt

# Replace this list with your data
data = [250, 275, 310, 320, 350, 365, 410, 415, 430, 500, 525, 610, 720, 840]

# Choose the number of bins (try changing this!)
plt.hist(data, bins=8)

plt.title("Histogram")
plt.xlabel("Value")
plt.ylabel("Frequency")
plt.show()
```

Try changing the value of `bins`. Notice that the overall shape can look different depending on how you group the data.

How do we compare dotplots, stemplots, and histograms?

For small data sets, dotplots and stemplots can be better because they show the exact values. In a histogram, you lose that detail because values are grouped into bins. In exchange, histograms make it easier to see the overall pattern in large data sets.

1.3.7 Cumulative Frequency Plot

A *cumulative frequency plot* (also called a *cumulative frequency chart*) shows the running total of how many data values are at or below each value (or at or below each class boundary). Instead of showing how many observations fall in each bin like a histogram, a cumulative frequency plot shows how the totals add up as you move from left to right.

Cumulative frequency plots are useful when you want to answer questions like:

- How many observations are less than or equal to a certain value?
- How many observations are greater than or equal to a certain value?
- What percent of the data fall below a certain cutoff?

How to Make a Cumulative Frequency Plot

1. Draw the x-axis and y-axis.
2. Set up bins for the data and identify the upper class boundaries (the right endpoints).
3. Compute the cumulative frequency at each upper class boundary.
   (This means: add the frequency of that class to the total from all previous bins.)
4. Plot a point at each upper class boundary with height equal to the cumulative frequency.
5. Connect the points with straight line segments.

Interpreting the shape of your plot

The steepness of the curve tells you where the data values are concentrated.

- If the curve rises quickly over a certain x-interval, many observations fall in that range.
- If the curve is almost flat, few observations fall in that range.

A cumulative frequency plot can also hint at skewness:

- **Right-skewed:** the curve increases quickly at smaller values, then levels off later.
- **Left-skewed:** the curve increases slowly at first, then rises sharply near the end.
- **Symmetric:** the curve often looks S-shaped.

Example

Suppose $n = 80$ observations were grouped into bins, and the cumulative frequency at $x = 350$ is about 23. Then:

- About 23 observations are ≤ 350.
- About $80 - 23 = 57$ observations are ≥ 350.

The plot below shows a sample cumulative frequency curve. (These coordinates are an example format; you can replace them with your own cumulative totals.)

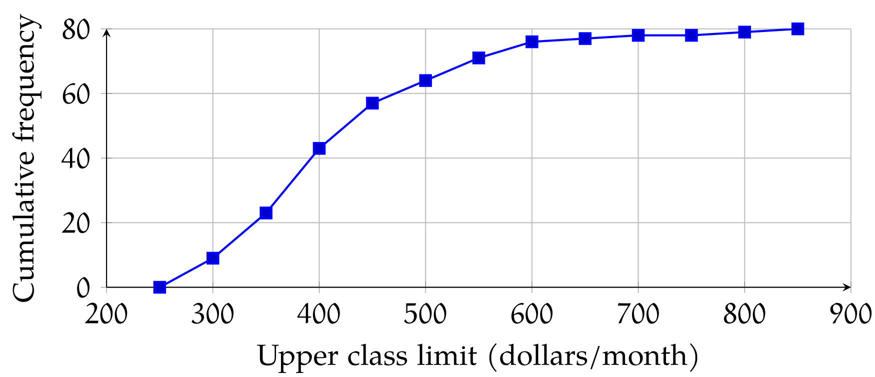

Figure 1.10: A cumulative frequency plot.

Making a Cumulative Frequency Plot in Python

This example shows the basic idea: build bins, count how many values fall in each bin, then take a cumulative sum.

```
import numpy as np
import matplotlib.pyplot as plt

# Replace with your data
data = np.array([250, 275, 310, 320, 350, 365, 410, 415, 430, 500, 525, 610, 720, 840])

# Choose bin edges (upper class boundaries are the right endpoints)
```

```
bins = np.arange(250, 851, 50)  # 250, 300, ..., 850

# Frequency in each bin
counts, edges = np.histogram(data, bins=bins)

# Cumulative frequency at each upper boundary
cumulative_counts = np.cumsum(counts)
upper_bounds = edges[1:]  # right endpoints

plt.plot(upper_bounds, cumulative_counts, marker='o')
plt.title("Cumulative Frequency Plot")
plt.xlabel("Upper class boundary ($)")
plt.ylabel("Cumulative frequency")
plt.grid(True)
plt.show()
```

If you change the `bin` width, the curve may look different, but it will always increase from left to right and end at n.

Boxplot

1.4 Bivariate Data

1.4.1 Scatter Plot

Sometimes, you have a large number of data points with two values, and you are looking for a relationship between them. For example, maybe you write down the average temperature and the total sales for your lemonade stand on the 15th of every month:

Date	Avg. Temp.	Total Sales
15 January 2022	2.6º C	$183.85
15 February 2022	-4.2º C	$173.56
15 March 2022	13.3º C	$195.22
15 April 2022	26.2º C	$207.61
15 May 2022	27.5º C	$210.88
15 June 2022	31.3º C	$214.18
15 July 2022	33.5º C	$215.23
15 Aug 2022	41.7º C	$224.07
15 September 2022	20.7º C	$198.94
15 October 2022	17.2º C	$196.10
15 November 2022	1.7º C	$185.10
15 December 2022	0.2º C	$188.70

You may wonder, ”Do I sell more lemonade on hotter days?”

To figure this out, you might create a scatter plot. For each day, you put a mark that represents that temperature and the sales that day:

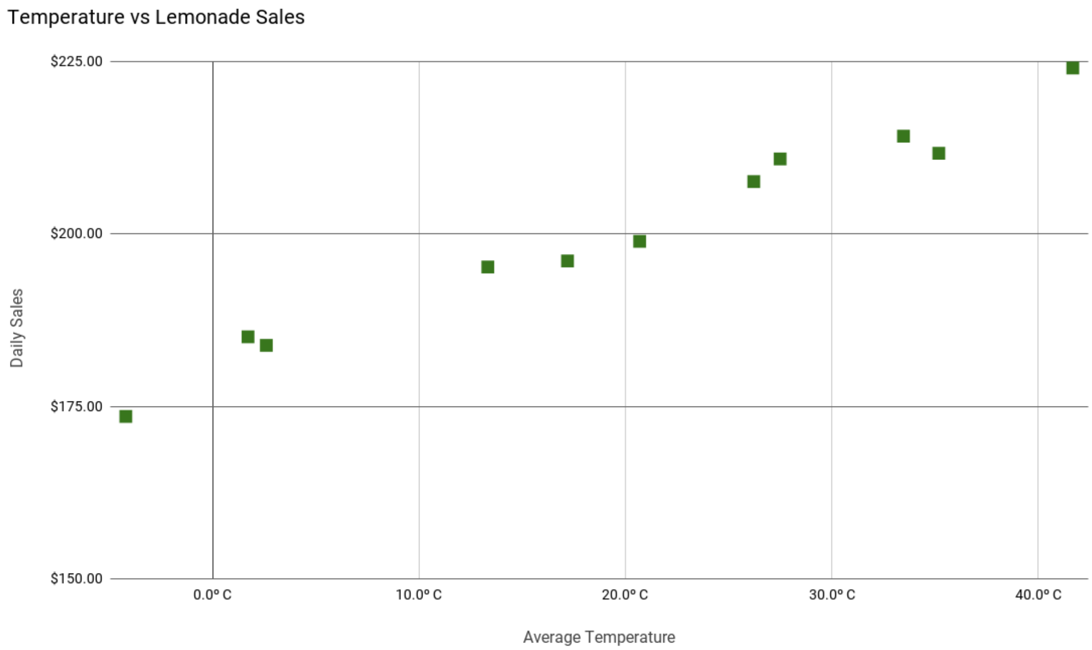

Figure 1.11: A scatterplot of temperature versus daily sales.

From this scatter plot, you can easily see that you do sell more lemonade as the temperature goes up. Drawing a *best-fit* line along the points will give you a *correlation coefficient*. A positive correlation coefficient will give you a positively proportional relationship, while a negative coefficient will give you a inversely proportional relationship.

Making a Scatter Plot in Python

```python
import matplotlib.pyplot as plt

# Avg. Temp (°C) and Total Sales ($)
temps = [2.6, -4.2, 13.3, 26.2, 27.5, 31.3, 33.5, 41.7, 20.7, 17.2, 1.7, 0.2]
sales = [183.85, 173.56, 195.22, 207.61, 210.88, 214.18, 215.23, 224.07, 198.94,
↪  196.10, 185.10, 188.70]

plt.scatter(temps, sales)
plt.title("Temperature vs. Lemonade Sales")
plt.xlabel("Average Temperature (°C)")
plt.ylabel("Total Sales ($)")
plt.grid(True)
plt.show()
```

Appendix A

Answers to Exercises

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